Endpoint – Academic Hub

Final Defense

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Software Engineering and Design Course
Constructor University, 2026

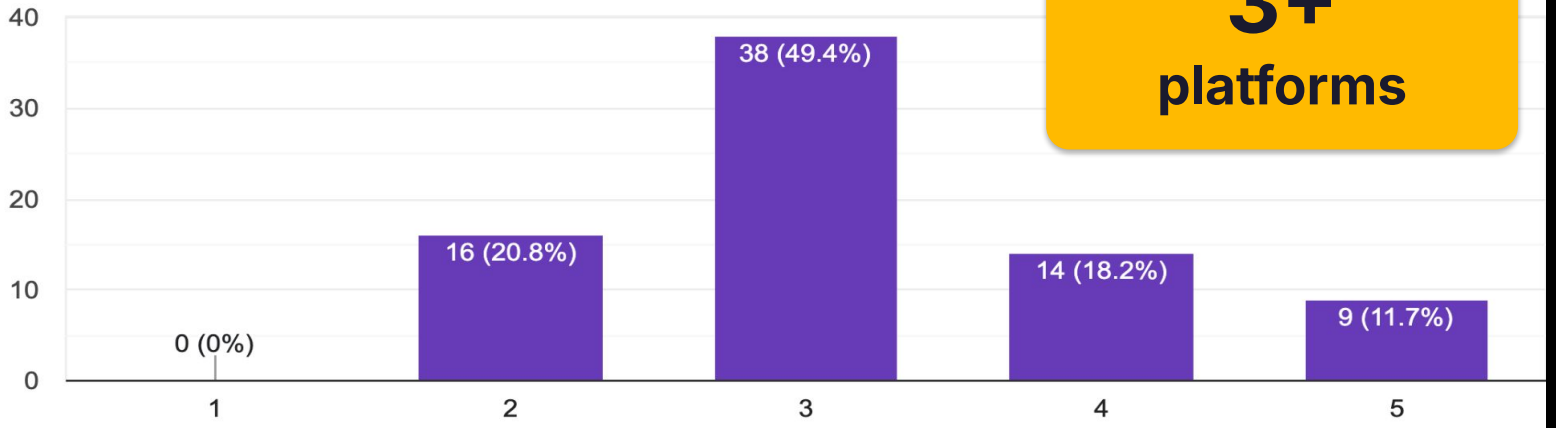
Product Value

Problem

Disperse Academic Information

Most students use at least 3 platforms for tracking courses

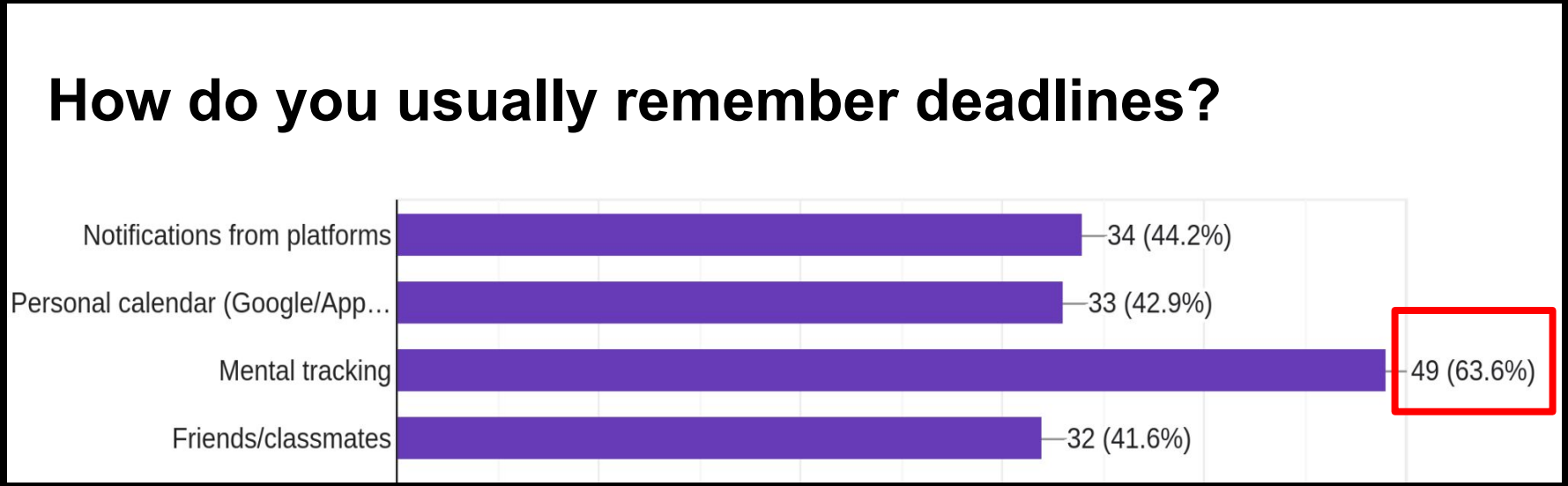
From how many different platforms do you receive academic information?



3+ platforms

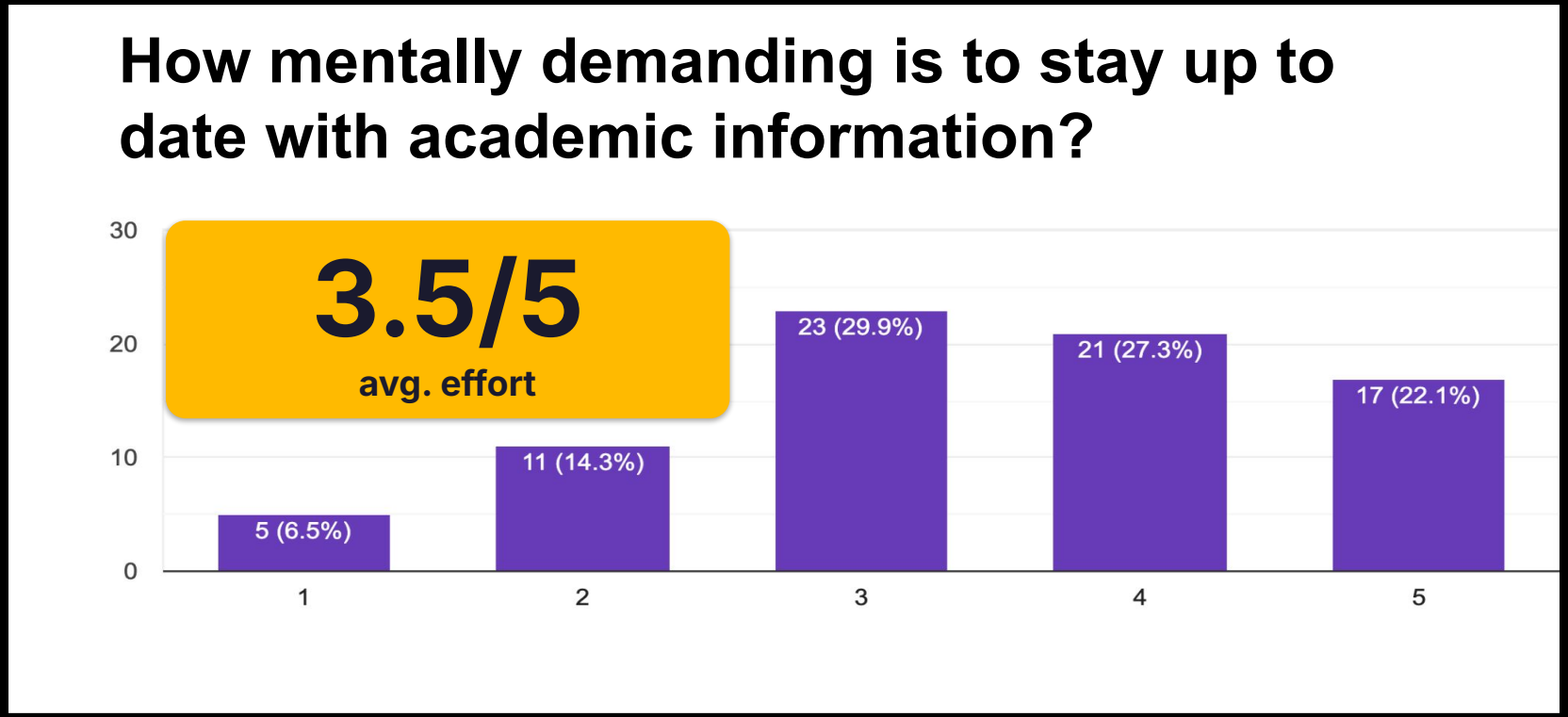
Cognitive Load

63% of students do mental tracking to remember their deadlines



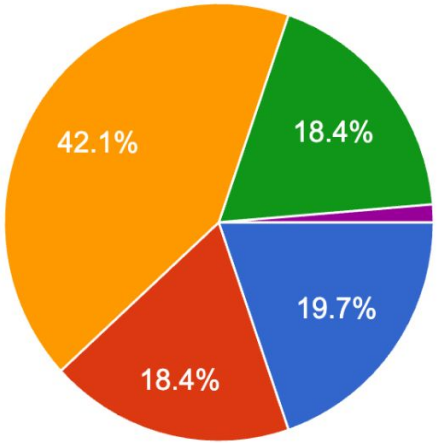
Cognitive Load

Staying up-to-date requires some effort



Missed Deadlines

How often have you missed a deadline because you did not see an update on time?



- Never
- Once
- A few times
- Many times
- Prefer not to say

79%

Missed deadlines

Solution

Endpoint – Academic Hub

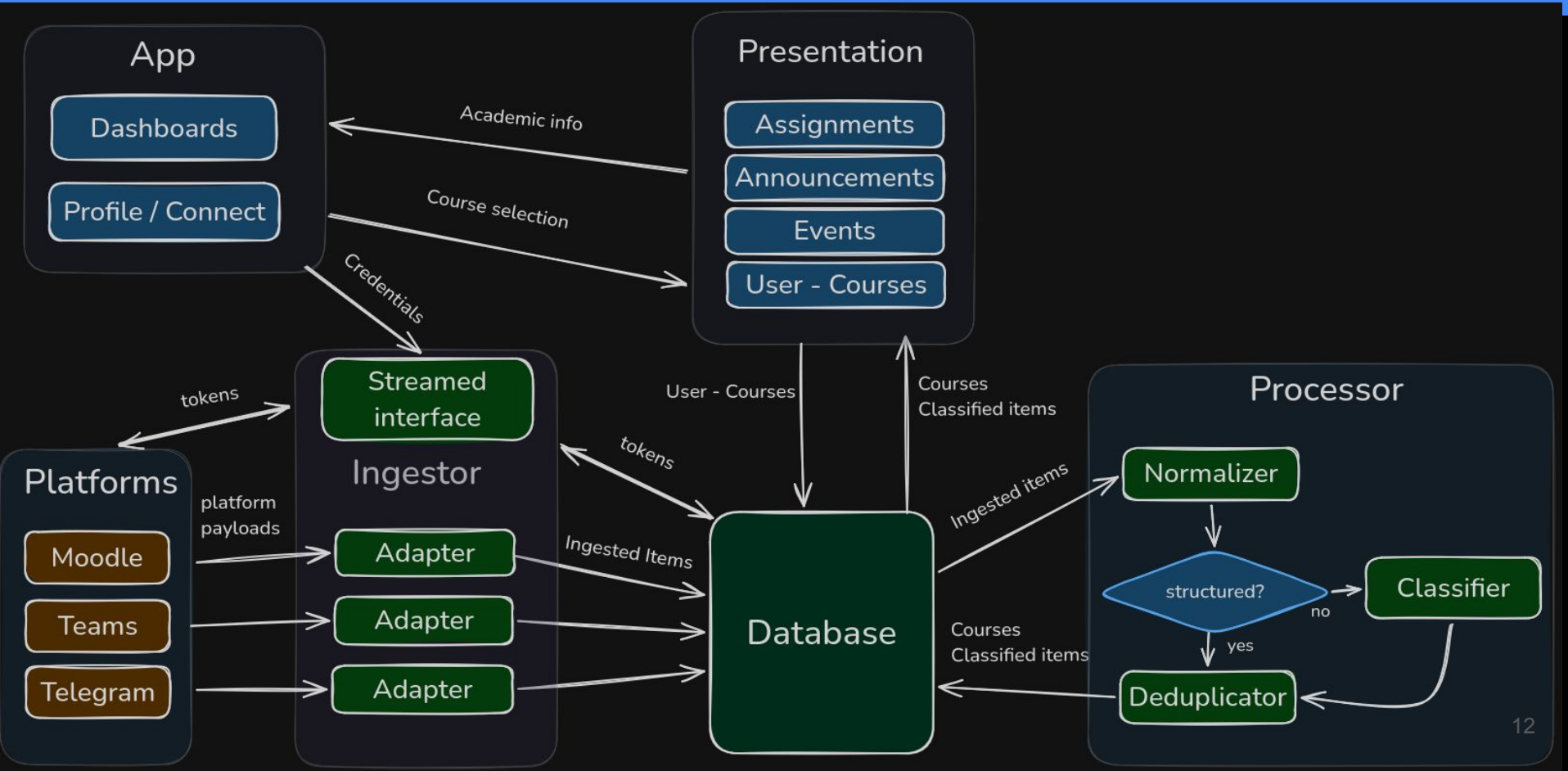
- Automatically updated academic information from all platforms
- Unified dashboard, one place to check
- Quick access to sources
- Easy tracking
  - Search
  - Filter
  - Organize

Benefits

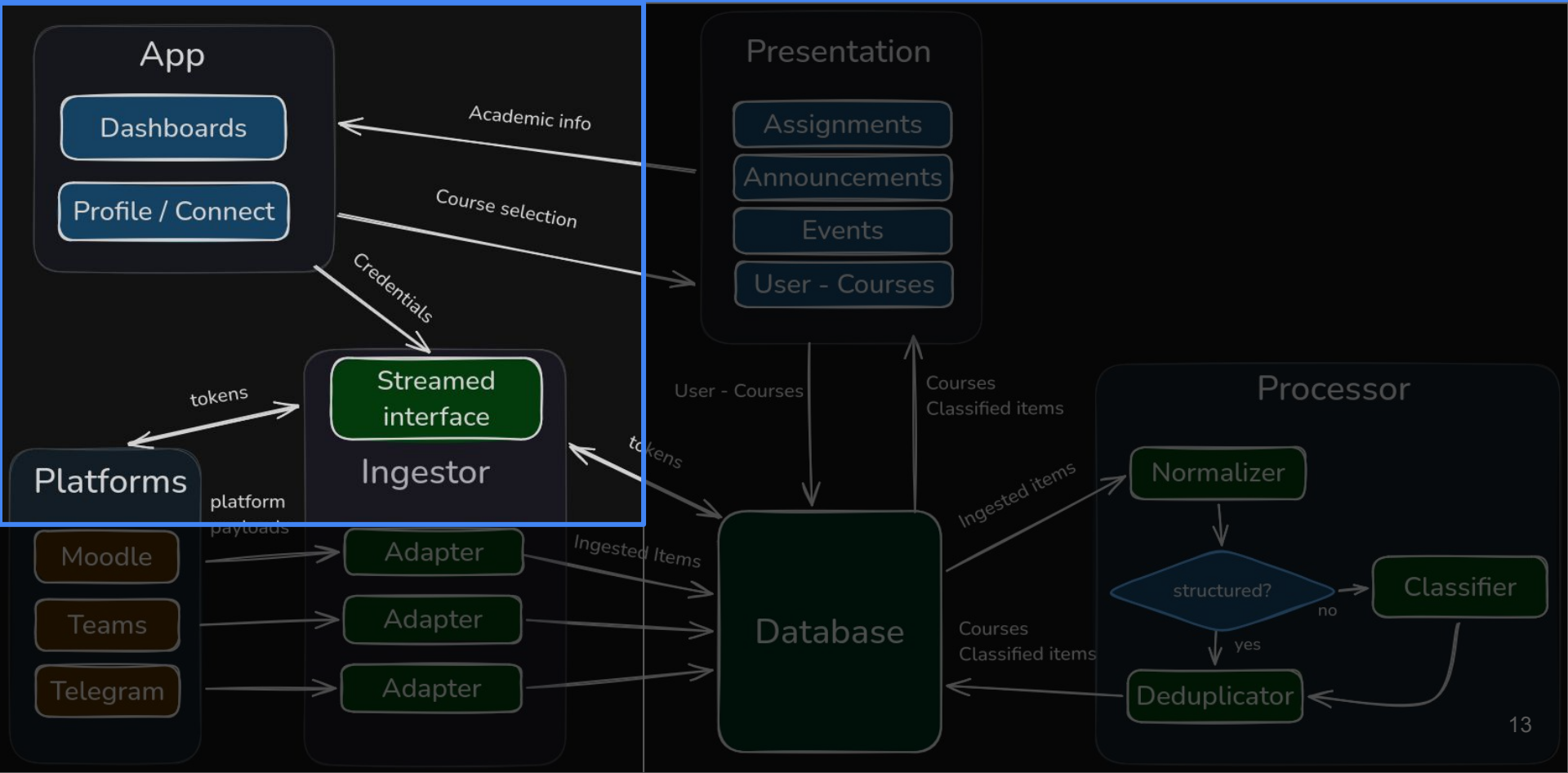
- Reduced tracking time and effort
- Visible recent updates and upcoming deadlines
- No missed announcements or study materials

Demo

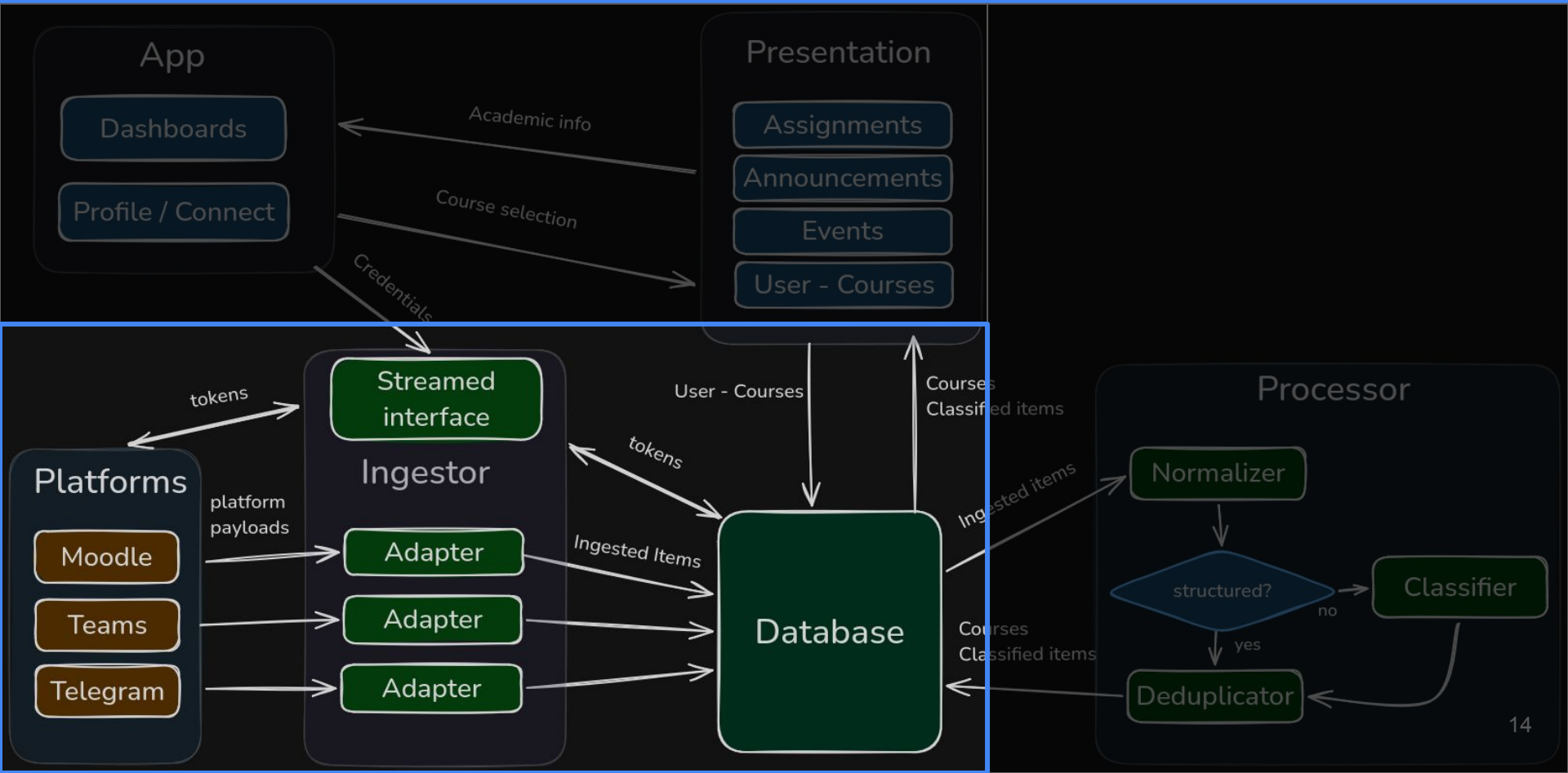
Data flow


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Profile / Connect
Academic info
Course selection
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Presentation
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Announcements
Events
User - Courses
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Streamed interface
Ingestor
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Classified items
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platform payloads
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Normalizer
structured?
no
Classifier
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Deduplicator
Courses
Classified items

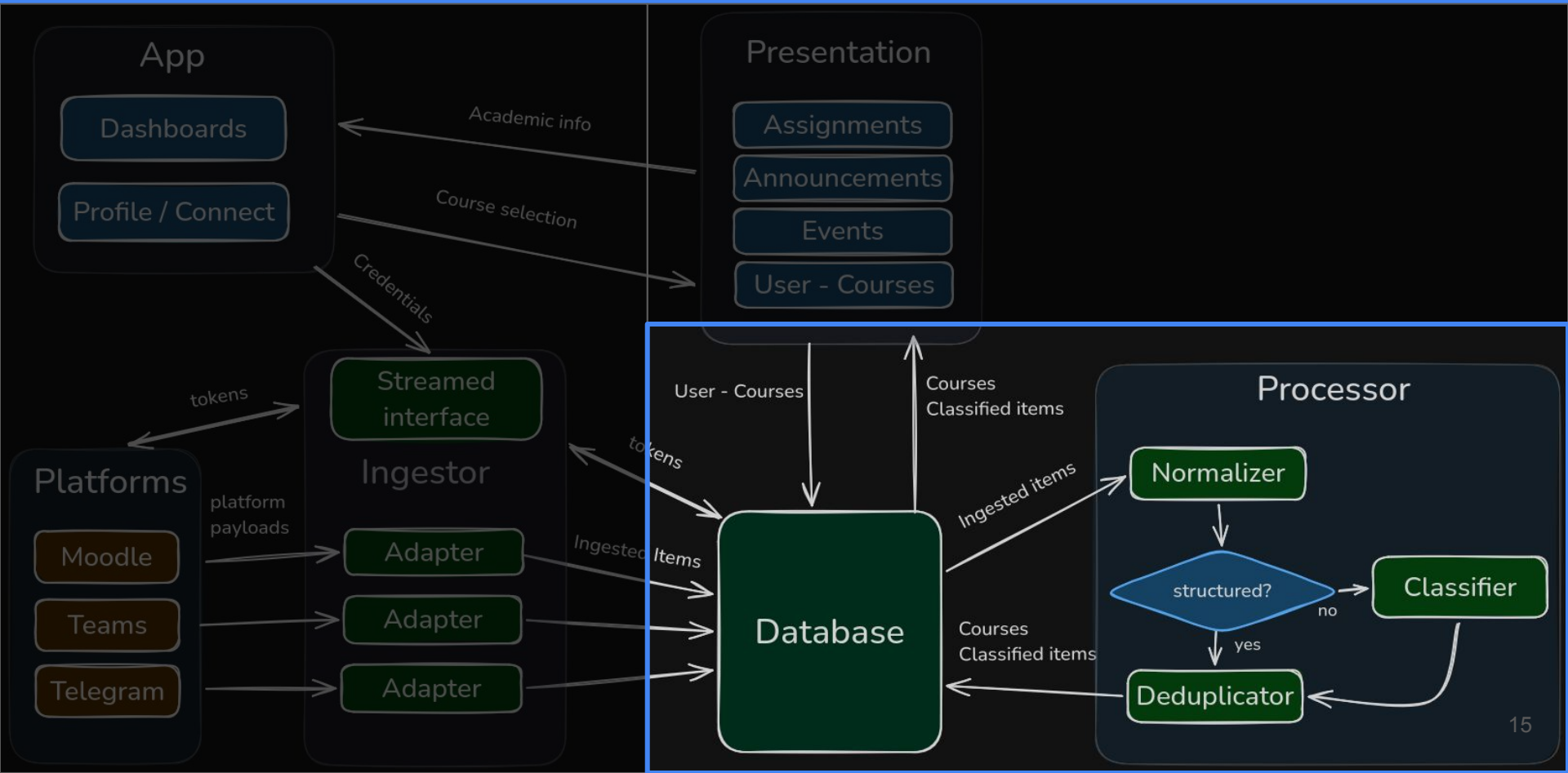
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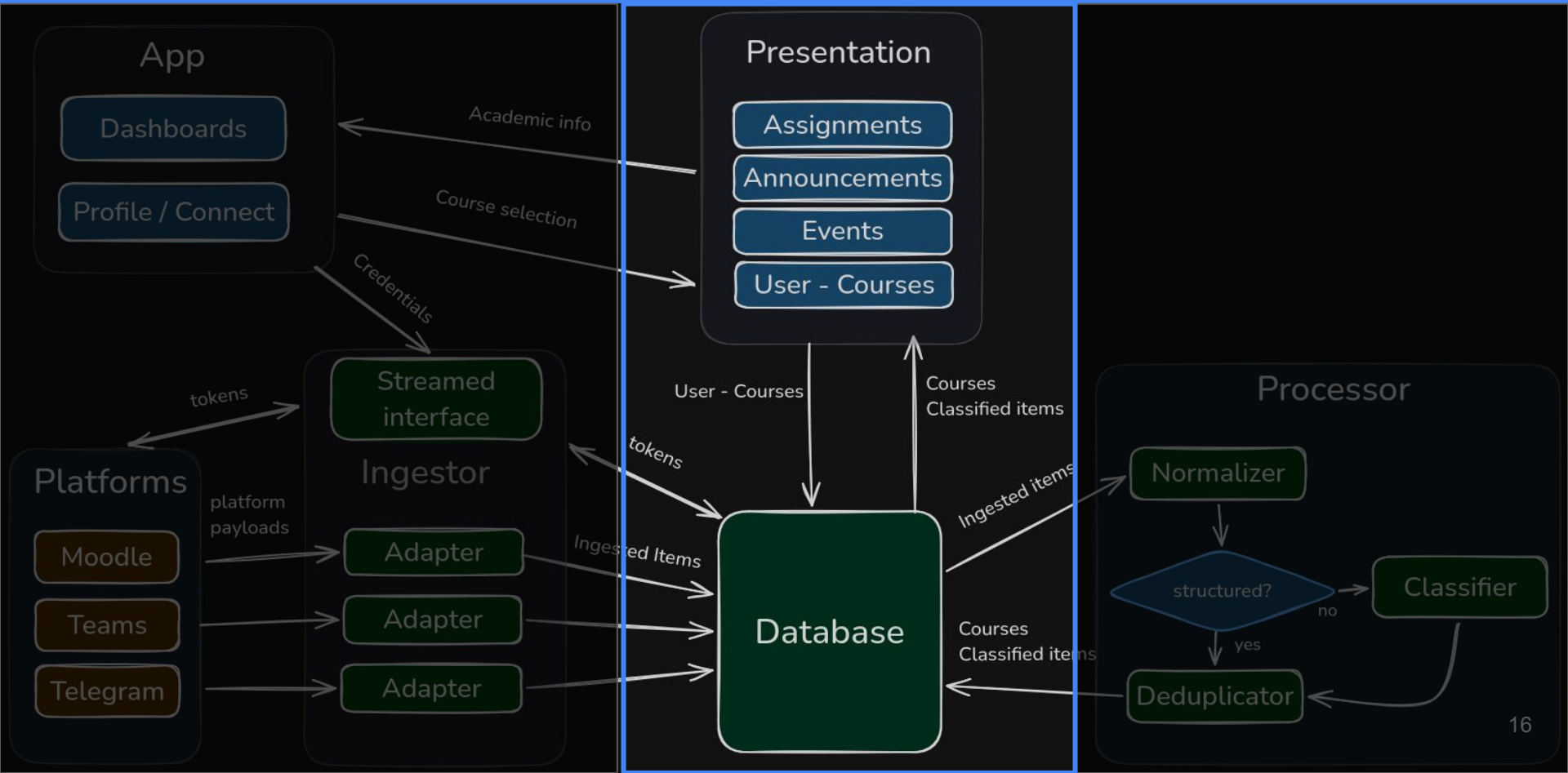
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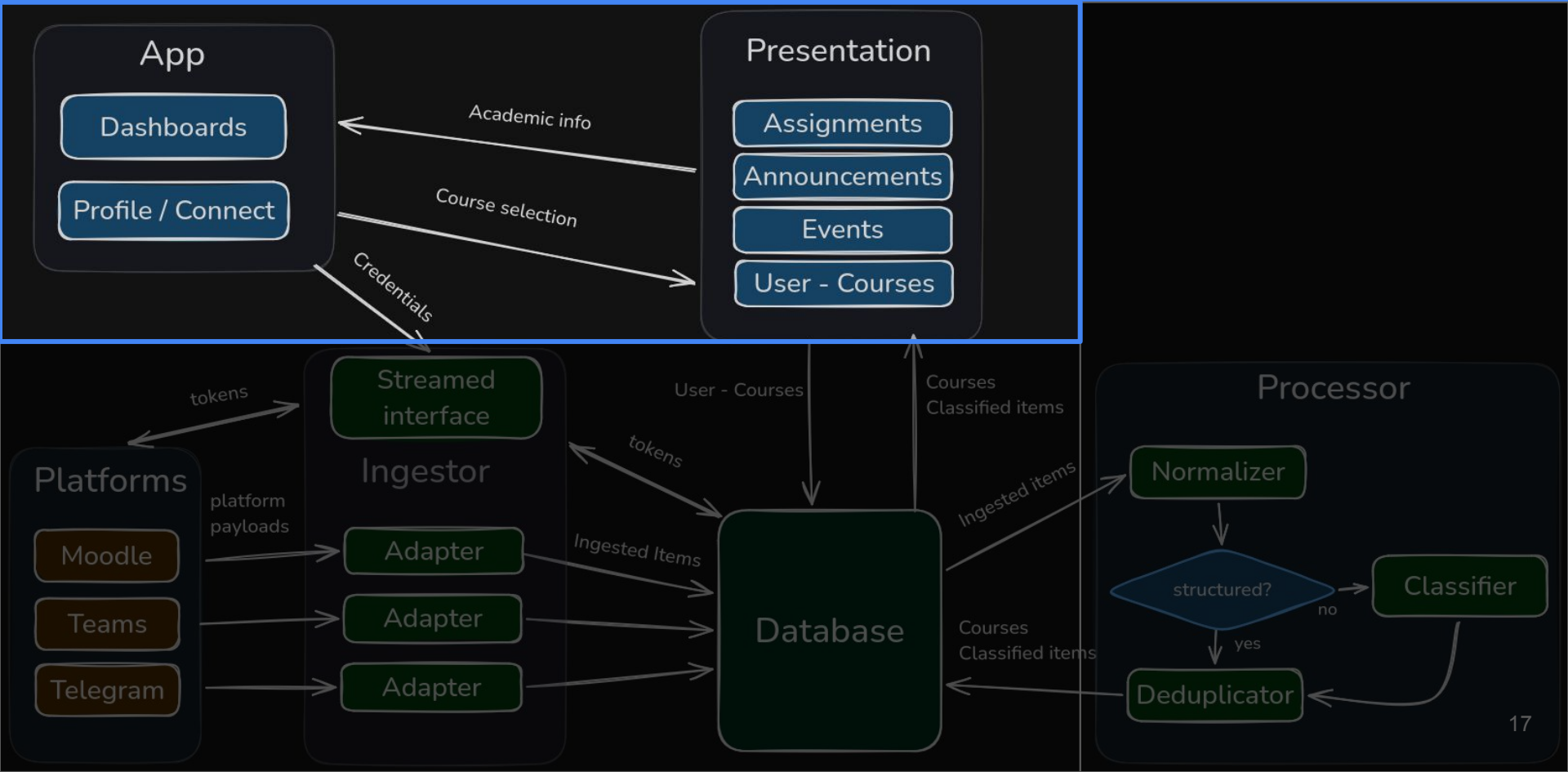
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Results

Ran against SDT and CS students accounts

Platform coverage

What can be fetched from the platforms?

Moodle

- Assignments
- Materials
- Calendar events
- Forums

Teams

- Assignments
- Channel Messages

Telegram

- Chat Messages

Performance

Credentials

- Teams auth token acquisition: <3 seconds after login
- Teams auth token refresh: 600 ms
- Moodle auth token acquisition: <1 second

Throughput under load

- Teams ingestion: 2,160 items/min
- Moodle ingestion: 540 items/min
- Processor: 135 items/min

Costs

- Average cost per LLM classification-extraction: $0.0015
- P90 peak-month unstructured messages per course: 128
- P90 monthly per course cost: $0.18

Evaluation

Dataset: 154 labeled unstructured messages · 17 courses · 2 platforms

Classification

- Micro F1: 0.82
- Macro F1: 0.83
- More details ahead

Extraction

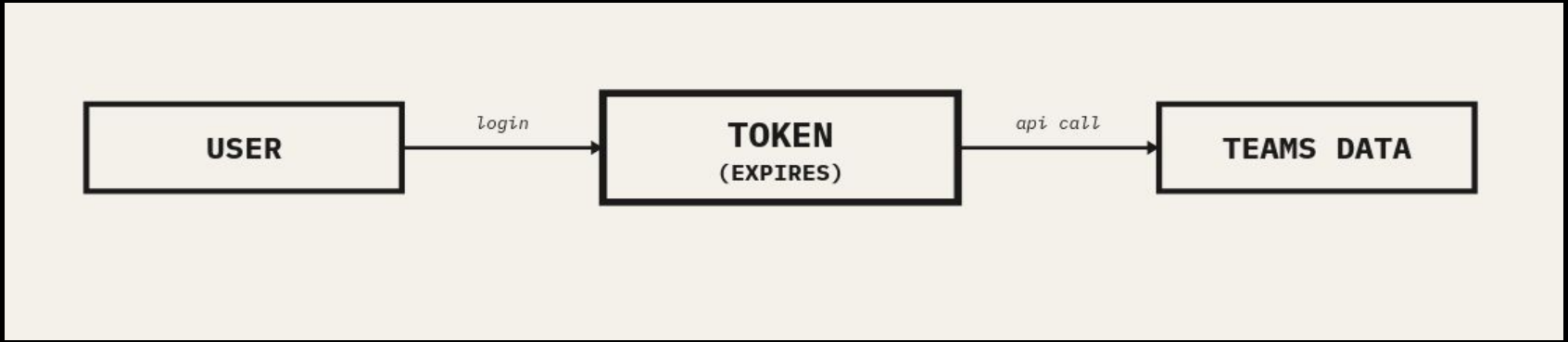
- NL Deadlines: 100%
- NL Event times: 100%
- Locations:
  - 76% exact
  - 95% similar

Technical Challenges

Auth Tokens

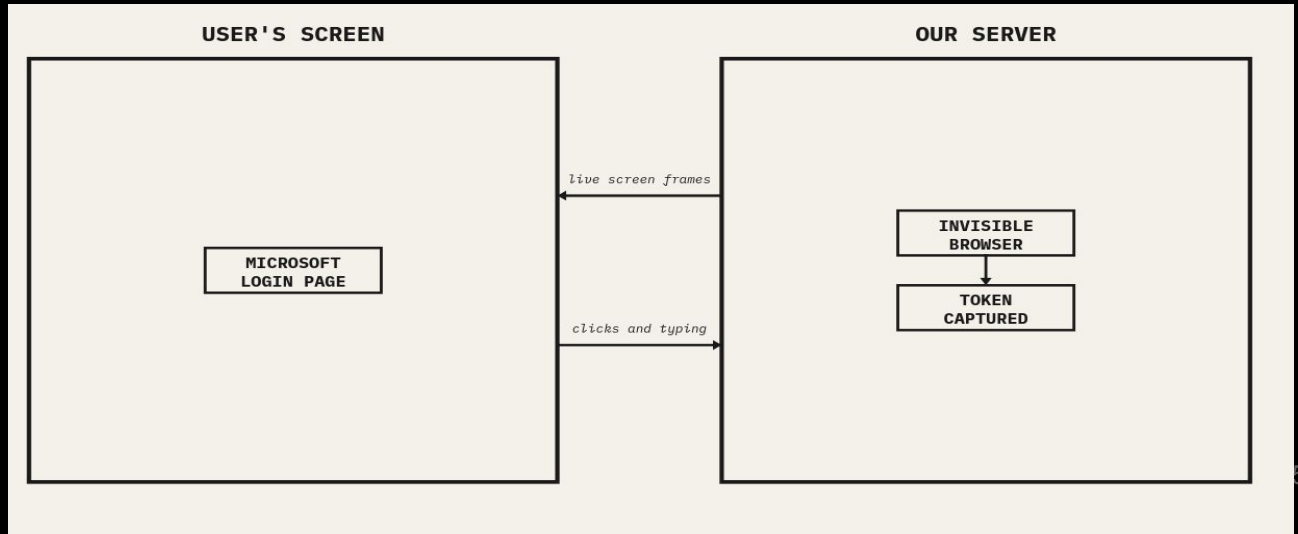
- Teams API calls need auth token. Bearer header, JWT, expires in about an hour
- User logs in → we capture auth token → lasts 90 days

- Two concerns: the first login, and keeping the token fresh.



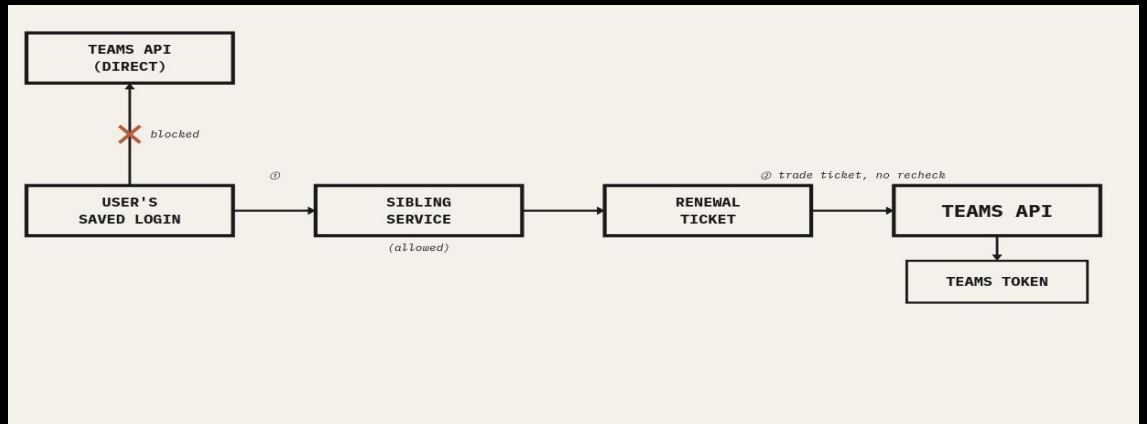
First login

- Microsoft login needs a real human at a keyboard. MFA!
- Log in on the user's machine and the token lives in their browser
  → out of our reach.
- We flip it. The browser runs on the server, the user sees it streamed. Same login,
  → token on our side.



Refresh

- Tried the naive approach with Playwright. Microsoft fingerprints automated browsers, the token comes back with the wrong permissions.
- Used Teams' own silent refresh path. Two HTTPS calls,removing the playwright dependency completely.
- From 30 seconds to 600 milliseconds fetch time.



The system

Wins:

- One login per student.
- 90 days of sync -> no daily auth.
- Refresh is cheap enough -> no rate-limits.

Signal in Chats

Information lives in course chats: Teams, Telegram

High Signal:

- Announcements
- Assignments
- Events
- Materials

Noise:

- Replies
- Questions
- Feedback polls
- Social chatter

Signal is buried → we must classify

Approaches

Some ideas:

- Heuristics: keyword occurrence, message length, URL presence, ...
- Train a linear classifier

Problems:

- Heuristics: fragile, hard to maintain
- Model training requires dataset with quality and volume

Let's use LLM

Prompt v1

You classify academic messages into one of five types:

- assignment
- announcement
- event
- material
- noise (use when the content doesn't match any other type)

Explain your reasoning briefly and choose the best type.

Prompt v1 Results

Micro F1: 0.77

Macro F1: 0.78

Announcement precision: 0.51

Noise recall: 0.61

expected \ predicted	ann	asg	evt	mat	raw
announcement	29	0	0	1	1
assignment	4	23	0	0	0
event	1	0	24	0	1
material	12	0	0	12	0
noise	11	0	0	0	17

There is almost no criteria → LLM defaults to announcement

Prompt v3.2

Category criteria

- Assignment: creates or updates required student work
- Event: attendable occurrence with time evidence
- Material: durable resources, links, slides
- Announcement: Course information with high signal score(not in above categories)
- Noise: default, no clear signal

Publishable content signals:

- Contains instruction, directive, durable information
- Addresses all group chat members
- Information is standalone, not context dependent

Prompt v3.2 → v4

- We tried using a scoring function on this criteria
- The model scored each signal in the range 0-2
- The model strongly scored weak similarities with provided criteria
- Lower announcement recall and noise recall

Changes

- All 3 signals must be present
- If there are mixed categories in message -> default to announcement

Prompt v4 Result

Micro F1: 0.82

Macro F1: 0.83

Announcement recall: 0.75

Noise recall: 0.75

Announcement precision: 0.67

expected \ predicted	ann	asg	evt	mat	raw
announcement	28	3	1	2	3
assignment	3	21	0	0	0
event	0	0	24	0	0
material	1	1	0	15	4
noise	10	0	1	1	36

Summary

Problem

- Students use 3+ academic platforms
- Effort to stay up to date
- Missed deadlines

Solution

- Unified dashboard, one place to check
- Auto-ingestion & LLM classification
- organized search & filter

Results

- 3 platforms, real student data
- Ingestion: 2,160 msg/min (Teams)
- Processor: 135 items/min
- Classification F1: 0.82 / 0.83
- LLM cost: ~$0.18/course/month

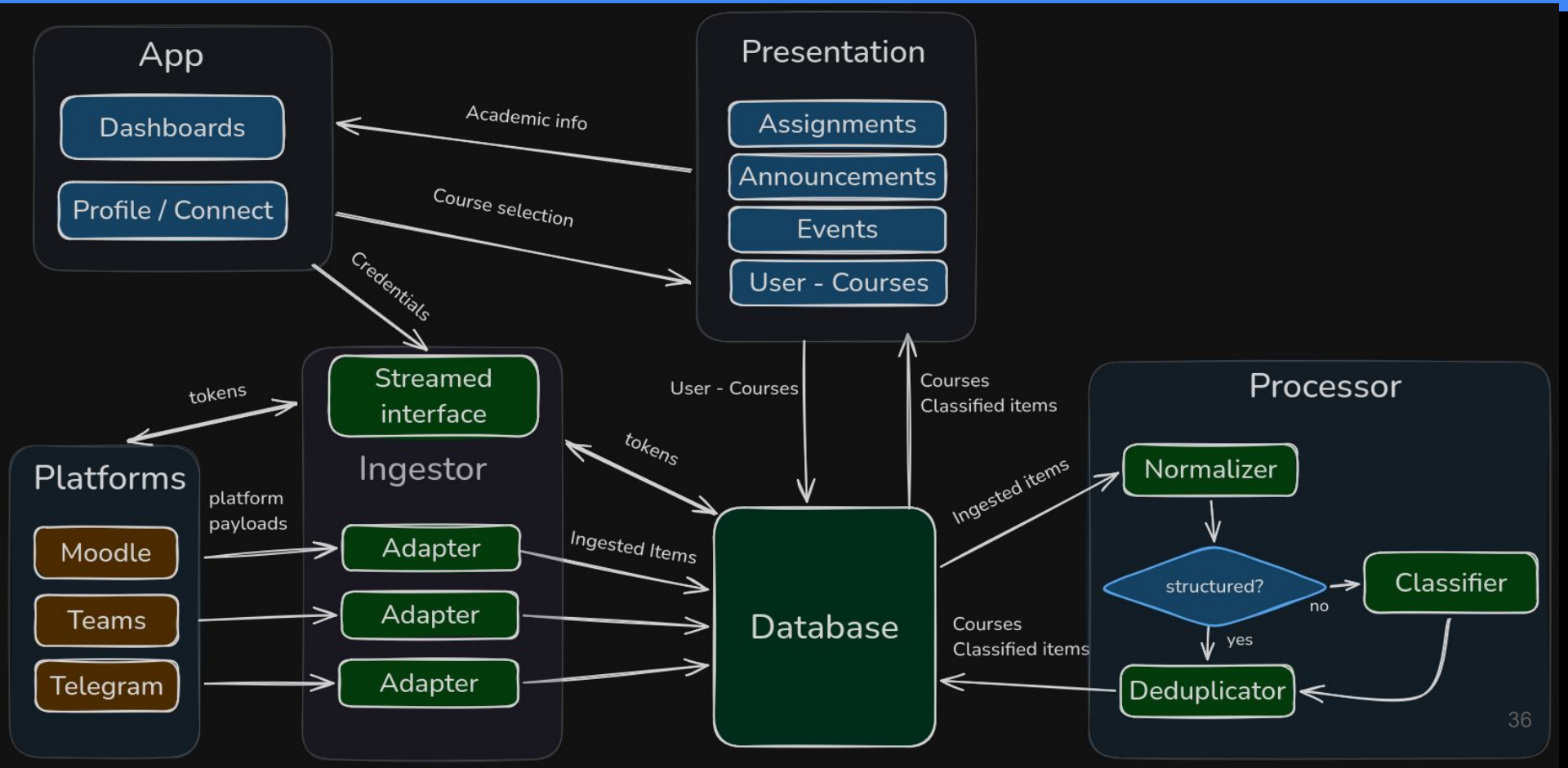
Teams auth

- Server-side browser streams MFA to user
- Silent token refresh: 600 ms, no browser
- One login, 90 days of uninterrupted sync

Classification

- LLM classifies signal from noise in chat messages
- Criteria-based scoring to reduce false positives
- Iterative prompt engineering, 4 versions
- Announcement precision 0.51 → 0.67
- noise recall 0.61 → 0.75

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Extra slides

Processing pipeline

FETCH BATCH → PROCESS → DEDUP → PERSIST

Concurrent processing:

- Items processed in async loop
- LLM calls are async - concurrency 80
- DB reads and writes are sync and batched

Batched persistence:

- One transaction per batch: classified items, canonical links, logs
- Trade-off: wait until batch finish -> low responsiveness under load,
- Throughput exceeds expected load -> tradeoff is acceptable
- If one row fails during batch write, fallback to per-item persist to isolate the error

Deduplication

Dedup pipeline:

- Group by course + entity type
- Load existing canonical representatives — DB + in-memory batch state
- Build Comparison objects
- Run match strategy — score candidates
- Run resolution strategy — decide action: create, link, or promote

Pluggable Strategies

Match strategy (WeightedMatchStrategy):

- Scores incoming item against canonical candidates
- Entity-specific field weights and thresholds

Resolution strategy:

- First Seen Resolution:
  - no match → create canonical
  - Match → link to existing item
- Platform Priority Resolution: promotes if incoming platform outranks current

Test Coverage

Used pytest-cov and vitest coverage

- Ingestor: 79%
- Processor: 89%
- Presentation: 69%
- Application: 40% - Main UI Happy Path

Prompt v3.2

Scoring Function:

- Score 3 signals: 0–2
- Score match with most similar category: 0-2

  total_score = (signal_1 + signal_2 + signal_3 + category_match) / 4

- If total_score >= 1.5 ⇒ publishable item

Prompt v3.2 Result

Micro F1: 0.71

Macro F1: 0.73

Announcement recall: 0.51

Noise recall: 0.56

Expected \ Predicted	ann	asg	evt	mat	raw
announcement	19	6	9	2	1
assignment	2	22	0	0	0
event	0	0	24	0	0
material	0	2	0	17	2
noise	14	0	6	1	27

Problems:

- Most generated scores are 2.
- Model strongly scores weak similarity with provided criteria

Feedback from students...

Too many websites used for information. Theres CampusNet, Moodle, and other websites and i have to check all when i need some specific information. It is inconvenient

Schedule updates can be unpredictable, but one does not receive instant updates about it

The most frustrating part is that information is spread across different platforms and it is hard to find everything in one place.

I missed my deadline for my presentation submission and it costed me my bonus points. It was because I had to look into 3 different websites to find the deadline

Always logging into several platforms a day.

Disperse Academic Information

